

DBSCAN — Worked Numerical

Classify every point as core, border or noise and form the clusters.

Given

		Parameters
Point	Coordinates (x, y)	ϵ (epsilon) = 2 — neighbourhood radius
P1	(1, 1)	MinPts = 3 — minimum points (including the point itself)
P2	(2, 2)	Distance: Euclidean
P3	(2, 1)	$d(A, B) = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$
P4	(8, 8)	
P5	(8, 9)	
P6	(25, 80)	

Definitions to apply

- **ϵ -neighbourhood $N_\epsilon(p)$** — all points within distance ϵ of p , including p itself.
- **Core point** — $|N_\epsilon(p)| \geq \text{MinPts}$.
- **Border point** — not core, but lies inside the ϵ -neighbourhood of some core point.
- **Noise** — neither core nor border.

Step 1 — Pairwise Euclidean distances

Only distances $\leq \epsilon = 2$ matter for neighbourhoods; those cells are shaded.

	P1	P2	P3	P4	P5	P6
P1	—	1.41	1	9.90	10.63	82.57
P2	1.41	—	1	8.49	9.22	81.32
P3	1	1	—	9.22	10	82.28
P4	9.90	8.49	9.22	—	1	73.98
P5	10.63	9.22	10	1	—	73.01
P6	82.57	81.32	82.28	73.98	73.01	—

Working for the four pairs that fall within ϵ :

$$d(P1, P2) = \sqrt{[(2-1)^2 + (2-1)^2]} = \sqrt{[1 + 1]} = \sqrt{2} = 1.41 \leq 2 \rightarrow \text{neighbours}$$

$$d(P1, P3) = \sqrt{[(2-1)^2 + (1-1)^2]} = \sqrt{[1 + 0]} = \sqrt{1} = 1 \leq 2 \rightarrow \text{neighbours}$$

$$d(P2, P3) = \sqrt{[(2-2)^2 + (1-2)^2]} = \sqrt{[0 + 1]} = \sqrt{1} = 1 \leq 2 \rightarrow \text{neighbours}$$

$$d(P4, P5) = \sqrt{[(8-8)^2 + (9-8)^2]} = \sqrt{[0 + 1]} = \sqrt{1} = 1 \leq 2 \rightarrow \text{neighbours}$$

Every other pair (e.g. $d(P3, P4) = \sqrt{[36 + 49]} = \sqrt{85} \approx 9.22$, $d(P5, P6) \approx 73.01$) is far larger than 2.

Step 2 — ϵ -neighbourhoods and point roles

Point	Neighbours within $\epsilon = 2$	$ N_\epsilon $ incl. itself	$ N_\epsilon \geq \text{MinPts} ?$	Role
P1	P2, P3	3	Yes ($3 \geq 3$)	Core
P2	P1, P3	3	Yes ($3 \geq 3$)	Core
P3	P1, P2	3	Yes ($3 \geq 3$)	Core
P4	P5	2	No ($2 < 3$)	Noise
P5	P4	2	No ($2 < 3$)	Noise
P6	none	1	No ($1 < 3$)	Noise

Why P4 and P5 are noise, not border. P4 and P5 are within ϵ of each other, but each has only 2 points in its neighbourhood (itself + the other), and $2 < \text{MinPts} = 3$, so neither is core. A border point must sit inside the ϵ -circle of a core point — and there is no core point anywhere near P4 or P5 (the nearest core point, P3, is 9.22 away). With nobody able to reach them, both are noise. Two points closer than ϵ do not make a cluster on their own; only a core point can start one.

Step 3 — Grow the clusters from the core points

1. Start at P1 (core). Open Cluster 1 = {P1} and queue its neighbours P2, P3.
2. P2 is core ($|N_\epsilon| = 3$) → joins Cluster 1 and its neighbours (P1, P3) are queued — both already handled.
3. P3 is core ($|N_\epsilon| = 3$) → joins Cluster 1. Queue empty → Cluster 1 = {P1, P2, P3} is complete.
4. P4: not core, not reached by any core point → noise. P5: same → noise. P6: no neighbours at all → noise.
5. No unvisited core point remains, so no further cluster can be started.

Final answer

Result	Points	Reason
Cluster 1	P1 (1,1), P2 (2,2), P3 (2,1)	all three are core points
Noise	P4 (8,8), P5 (8,9), P6 (25,80)	P4/P5: a pair below MinPts; P6: isolated

Number of clusters = 1; core = 3, border = 0, noise = 3.

Check the parameters — would a small change alter the answer?

DBSCAN's answer depends on $(\epsilon, \text{MinPts})$; a quick sensitivity check shows how firmly this result stands.

Setting	Clusters	Noise	What changed
$\epsilon = 2, \text{MinPts} = 3$ (given)	{P1, P2, P3}	P4, P5, P6	the result above
$\epsilon = 2, \text{MinPts} = 2$	{P1, P2, P3}, {P4, P5}	P6	P4 and P5 now reach MinPts, so a second cluster appears
$\epsilon = 1, \text{MinPts} = 3$	{P1, P2, P3}	P4, P5, P6	P1–P2 (1.41) is lost but P3 (distance 1 to both) still links them
$\epsilon = 10, \text{MinPts} = 3$	{P1, P2, P3, P4, P5}	P6	P3 reaches P4 (9.22) and P5 (10.00), so the two groups merge
$\epsilon = 2, \text{MinPts} = 4$	none	all six	no point has 4 neighbours — everything is noise

Reading: the cluster {P1, P2, P3} survives for every ϵ between 1 and about 8, so it is a stable structure. Whether {P4, P5} counts as a cluster is decided entirely by MinPts — with MinPts = 3 the standard DBSCAN definitions make them noise. Compare this table with the k-distance plot and stability map in the DBSCAN interactive page.